Ronit Anilkumar

(707) 236-4722 | ranilkumar@ucla.edu | linkedin.com/in/ronit-anilkumar | ronitanilkumar.com

Education

University of California, Los Angeles (UCLA) Los Angeles, CA
Bachelor of Science in Computer Science and Engineering *Expected June 2027*

Coursework

- Computer Systems Architecture
- Distributed Systems
- Computer Network Fundamentals
- Operating Systems

Experience

Machine Learning Researcher | UCLA Biomedical AI Research Lab Nov. 2024 – Present
University of California, Los Angeles *Los Angeles, CA*

- Co-authored US-JEPA, a self-supervised foundation model for ultrasound cancer detection trained on **4.73M** frames across 49 datasets using **PyTorch** on A100 GPU clusters, achieving **1.6x** throughput improvement through mixed-precision training and GPU profiling.
- Built fault-tolerant distributed data pipelines processing **4.73M+** ultrasound frames with automated quality validation, preprocessing workflows, and domain-specific augmentation across **8+** anatomical domains.
- Engineered end-to-end computer vision evaluation pipeline implementing custom multi-task sampling, domain-specific loss functions (Dice, IoU), and comprehensive metrics (AUC, F1, Hausdorff Distance) across diverse clinical tasks.

Projects

Loom | GitHub | *Rust, Tokio, Axum, Reqwest, SHA-256* May 2026 – Present

- Built a parallel model weight streaming agent in **Rust** targeting GPU cold-start latency, splitting files into **10 MB** chunks fetched concurrently via HTTP Range requests with a semaphore-bounded worker pool and exponential backoff retry, reducing multi-gigabyte model load times versus sequential object storage fetches.
- Implemented a content-addressable disk cache keyed on **SHA-256** URL hashes with hard-link assembly, making cache hits nearly instantaneous regardless of file size by eliminating data copies entirely.

Veylor | GitHub | *LangGraph, Python, FastAPI, Docker, OpenTelemetry, Semgrep* June 2025 – Present

- Designed and built an autonomous multi-agent CI/CD repair system using **LangGraph** orchestration with specialized agents for fault localization, patch generation, and test validation, enabling end-to-end automated repair without human intervention via **GitHub Actions** integration.
- Built security-hardened **Docker** sandboxes for isolated code execution with strict resource limits (512MB RAM, 1 CPU, 60s timeout), no-network policies, dropped Linux capabilities, and **Semgrep** CVSS v3.1 static analysis enforcing zero vulnerabilities across **1,000+** patches validated.
- Implemented **OpenTelemetry** distributed tracing across the full multi-agent pipeline with p50/p95/p99 latency tracking, real-time CPU and memory telemetry, and complete audit trails of all LLM decisions, alongside a **FastAPI** backend with **40+** REST endpoints and SSE streaming.

CoWrite | GitHub | *Yjs, TipTap, WebSockets, SQLite, TypeScript, Node.js, Anthropic API* March 2026 – Present

- Built a real-time collaborative rich text editor using **CRDT-based** sync (**Yjs**) and **WebSocket** broadcast with conflict-free concurrent edits, live cursor presence via the Awareness protocol, and offline-first reconciliation on reconnect.
- Designed a structured AI execution model where agents run against frozen document snapshots and produce typed operations (`replace_text`, `insert_block_after`), validated in two phases (server then client) before application, preventing stale or invalid edits under concurrent usage.

Technical Skills

Programming Languages: Python, Rust, C++, TypeScript, JavaScript, SQL
Systems & Databases: Distributed systems, CRDTs, WebSockets, FPGA, Verilog, PostgreSQL, Redis, SQLite
Cloud & Infrastructure: AWS, Docker, Linux, GitHub Actions, CI/CD, OpenTelemetry